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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
7TH SEMESTER, 2024-25

A VISION BASED DRIVER ATTENTION MONITORING SYSTEM

Presented by: Group 4

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INTRODUCTION

- Driver distraction is a major cause of unsafe driving, which leads to the severe vehicle accidents every year. Distracted driving includes eating snakes while driving, talking to passengers, texting on the phone, talking on phone, and feeling drowsiness.

- Identifying drivers behaviors is therefore becoming increasingly important for car manufacturers, our aim to build in-car intelligence that can improve safety by notifying drivers in real-time of potential hazards.

- We present a vision-based model for monitoring driver's distracted states using Deep Learning.

- The model will be trained to identify ten common driving-related tasks and also to determine whether the driver is being distracted or not.

- This project aims to develop a desktop application that will help the drivers and passengers to protect from accidents.

LITERATURE SURVEY

Author	Tittle	Year of publication	Method	Advantages/ benefits	Disadvantages
Anjali Sanjivanrao, Kalyani Manoj Swamy, et al.	Book Recommendation System Using Machine Learning	2022	Collaborative Filtering Algorithm, Support Vector Machine (SVM), and Hybrid Recommendation Engine	1. Cost - Efficient 2. Accurate 3. Reliable 4. User - friendly	1. New items 2. Limited variety in recommendation 3. Reliance on user interaction and feedback
Anoop A ,N Ayush Ubale	The Cloud Based Collaborative Filtering Algorithm for Library Book Recommendation System	2020	It utilizes a hybrid recommendation engine and collaborative filtering algorithm to recommend books based on user ratings, reducing time and cost	1. Efficient book selection 2. Reduce manual work	1. Reliance on user ratings and feedback for recommendations
Noor Ifada, Irvan Syachrudin et al.	The Cloud Based Collaborative Filtering Algorithm for Book Recommendation System	2020	It utilizes a hybrid recommendation engine and collaborative filtering algorithm to recommend books to users	1. Time and Cost-efficient 2. Reducing manual work for users	1. Rely heavily on user ratings and feedback 2. Potentially limiting the variety the variety of suggestions

LITERATURE SURVEY

Author	Tittle	Year of publication	Method	Advantages/ benefits	Disadvantages
HengSong Tan, HongWu Ye	A Collaborative Filtering Recommendation Algorithm Based on Item Classification	2019	The paper proposes a collaborative filtering recommendation algorithm based on item classification to address the sparsity problem in traditional collaborative filtering systems by pre-producing ratings through item classification and utilizing item-based collaborative filtering for recommendations	The proposed algorithm can alleviate the sparsity problem and provide better recommendations than traditional collaborative filtering	The item classification process adds complexity to the algorithm and may not be effective if the item attributes are not informative enough to accurately predict the missing ratings
Jaturawit Chaiwong, Nattapon Prajugjit, et al	Book Recommendation Website with Chatbot	2020	The method used in the Book Recommendation Website with Chatbot project is the integration of a chatbot named wannaRead, developed using Engati	Providing users with efficient book recommendations and answers to their questions	The chatbot's limitations in understanding complex or nuanced queries

PROBLEM STATEMENT

“To detect driver's distracted states and alert them to focus on driving by using proposed vision-based detection model that makes use of Deep Learning techniques.”

Input: Images / Videos for training and testing from webcam.

Output: Classification of different states and alerting the driver.

OBJECTIVES

- To develop a vision-based and real-time system that extracts driver’s attention features and classify them into various modes.
- To minimize number of accidents caused due to lack of driver's attention.
- To maintain a high alert state while driving a vehicle, alarm of distracted state is implemented using OpenCV and deep learning techniques.
- To implement RESNET based application to increase the accuracy in detecting driver.

REQUIREMENT SPECIFICATION

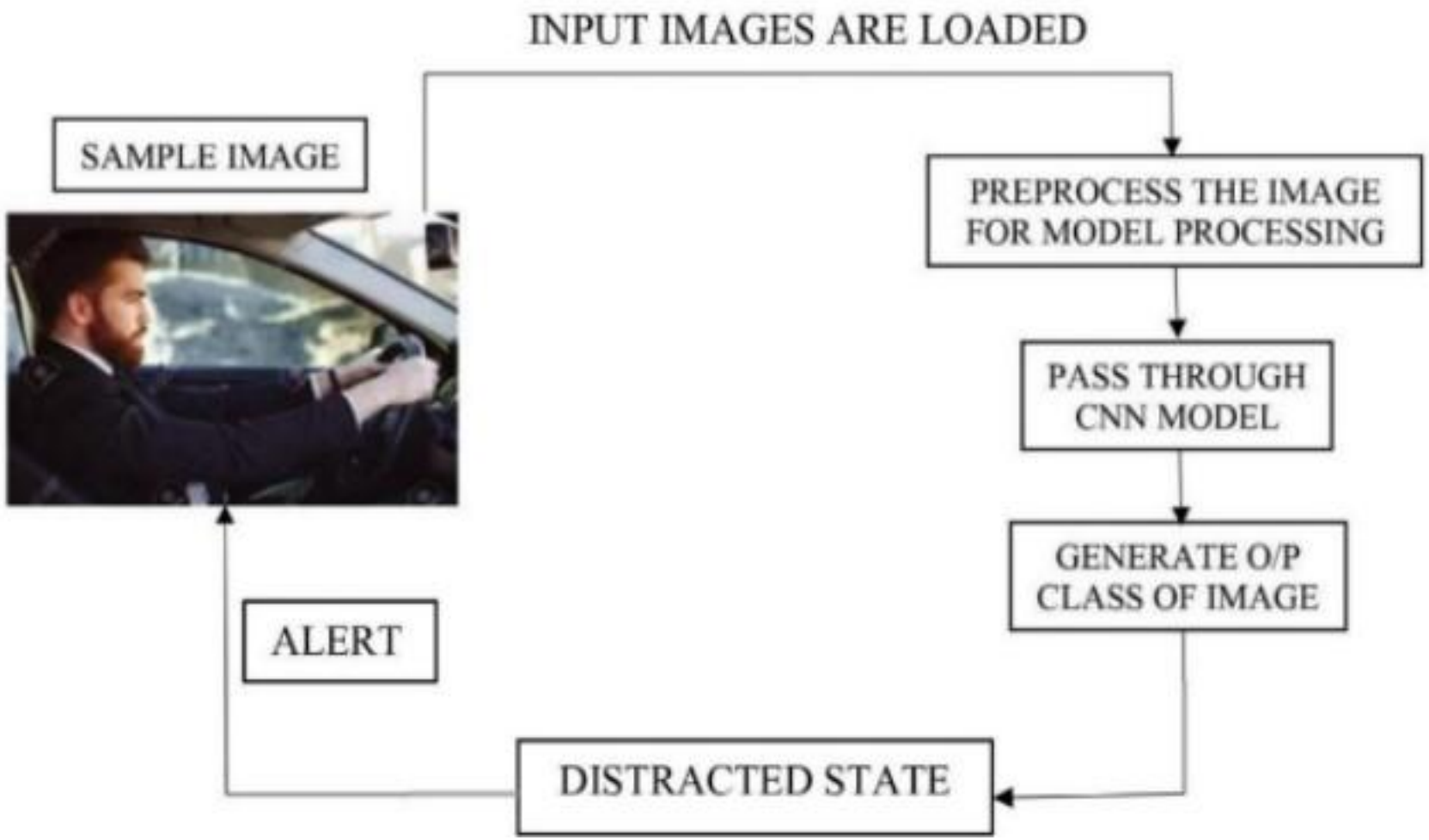
Hardware Requirements:

- **System Processor** : Core i3 / i5
- **Hard Disk** : 200 GB
- **Monitor :** 15’’ LED
- **RAM** : 8 GB
- Internet Broadband Connection with a speed of 2 Mbps or higher. ✓ Any desktop / Laptop system with above configuration or higher level

Software Requirements:

- **Operating system :**Windows 8/ 10/ 11
- **Programming Language :** Python 3.8
- **Framework :** PythonIDLE
- **Libraries Used :** tensorflow, torch, opencv-python, mediapipe, pillow, matpolib, keras, pandas and numpy

SYSTEM DESIGN



ACTIVITY DIAGRAM

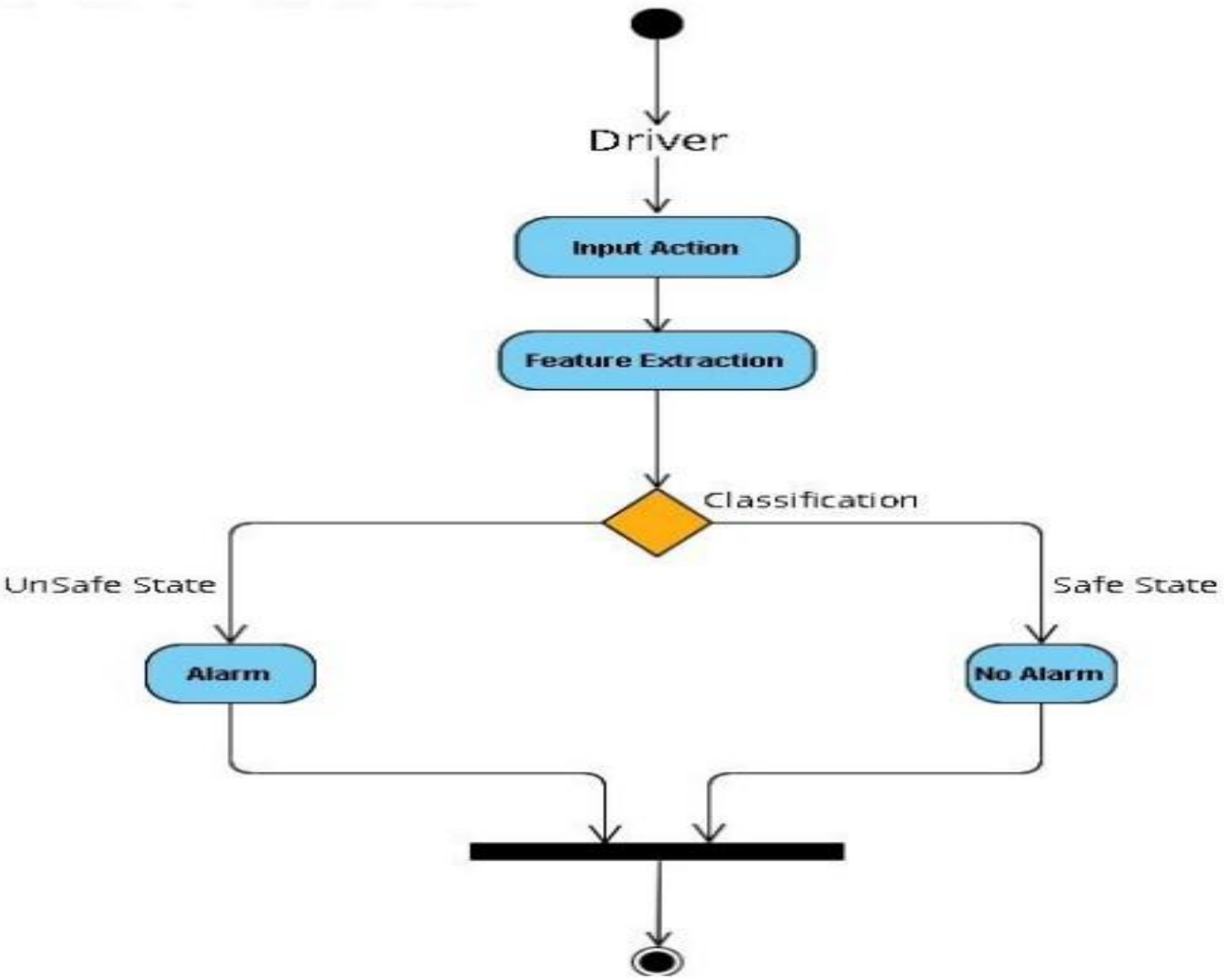


Figure 3.3: Activity Diagram

IMPLEMENTATION

Step 1 - Take Image as Input from a webcam.

- we will take images as input. So to access the webcam, we made an infinite loop that will capture each frame.
- We use the method provided by OpenCV, cv2.VideoCapture(0) to access the camera and set the capture object (cap).
- cap.read() will read each frame from web camera and we store the image in a frame variable.

Step 2 - Detect Face in the Image and Create a Region of Interest (ROI)

- To detect the face in the image, we need to first convert the image into RGB as the OpenCV algorithm for object detection takes rgb images in the input.
- We will be using haar cascade classifier to detect faces.

Step 3 - Detect the eyes from ROI and feed it to the classifier

- The same procedure to detect faces is used to detect eyes. First, we set the cascade classifier for eyes in left eye and right eye respectively then detect the eyes.
- Now we need to extract only the eyes data from the full image.
- This can be achieved by extracting the boundary box of the eye and then we can pull out the eye image from the frame.
- l_eye only contains the image data of the eye. This will be fed into our CNN classifier which will predict if eyes are open or closed. Similarly, we will be extracting the right eye into r_eye.

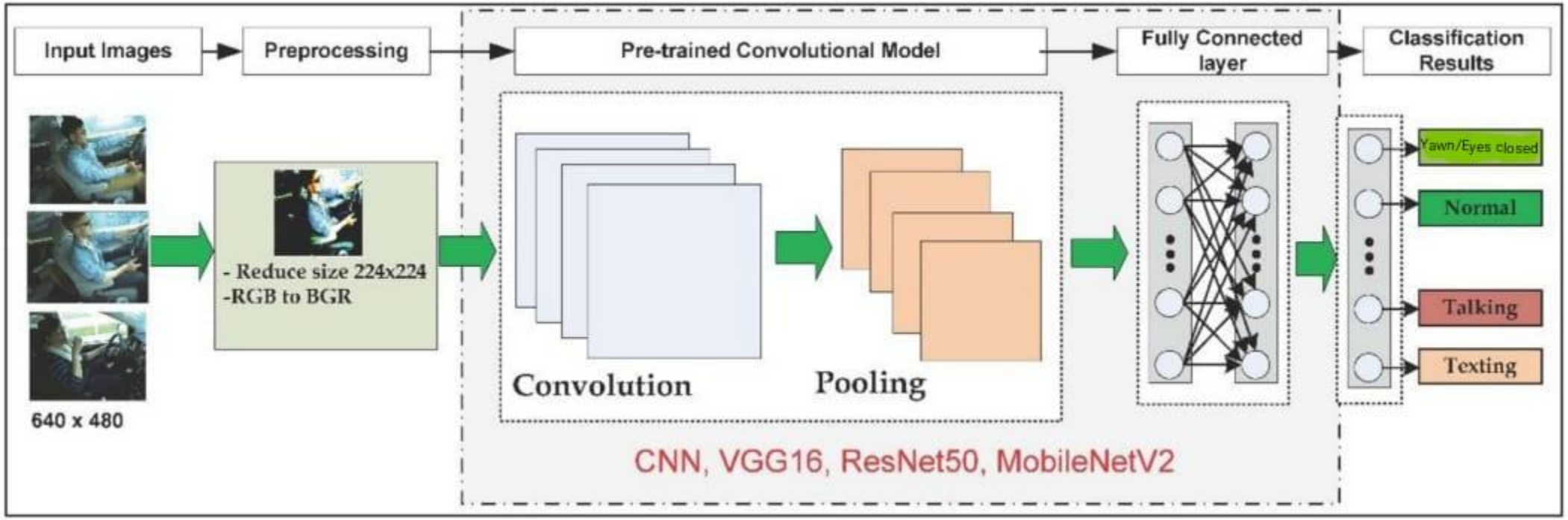
Step 4 - Classifier will Categorize whether Eyes are Open or Closed

- We are using CNN classifier for predicting the eye status. To feed our image into the model.
- First,RGB image is generated.
- Then, we resize the image to 24*24 pixels as our model was trained on 24*24 pixel images. We normalize our data for better convergence .
- Now we predict each eye with our model.
- If the value of Ipred[0] = 1, it states that eyes are open, if value of Ipred[0] = 0 then, it states that eyes are closed.

Step 5 - Calculate Score to Check whether Person is Drowsy.

- The score is basically a value we will use to determine how long the person has closed his eyes.
- So if both eyes are closed, we will keep on increasing score to alarm and when eyes are open, we decrease the score alarm stops.
- We are drawing the result on the screen using cv2.putText() function which will display real time status of the person.
- A threshold is defined for example if score becomes greater than 5 that means the person's eyes are closed for a long period of time. This is what we beep the alarm using sound.play().
- It uses a MobileNet model for driver action classification and a YOLO model for object detection.
- Actions with 0 reading will detect phone call actions, and reading 1 will detect texting actions and alarm rang.

Work-flow diagram of the proposal



RESULTS AND DISCUSSIONS

- The system proposes a method for detecting driver behavior using deep learning classification algorithms. In our proposed method, the CNN model provides high accuracy result, the evaluation metrics for multiclass classification including confusion matrix with probability of 0.99.
- On model training and testing the training accuracy is 0.9997, the validation accuracy is 0.9887, the training loss is 0.0003 and the validation loss found to be 0.0113. Therefore, the CNN model achieves 99% as overall accuracy score. Deep learning techniques achieve high accuracy and is effective as a new image classification technique.

SNAPSHOTS

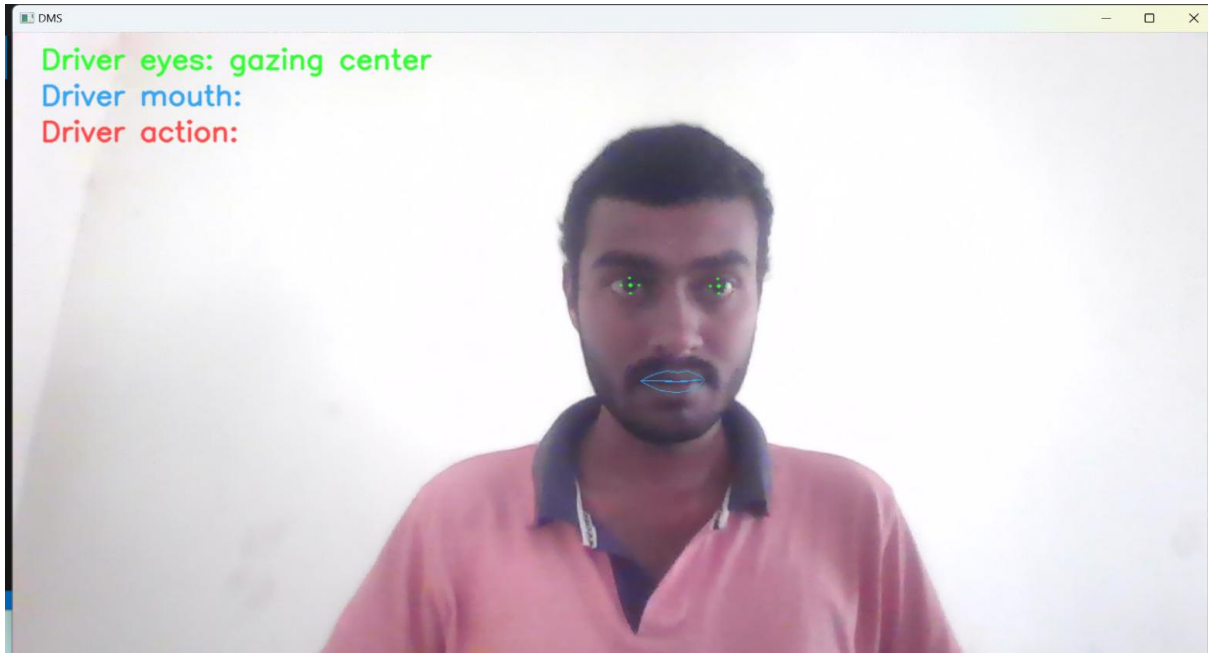


Fig 1: Safe Driving action Prediction

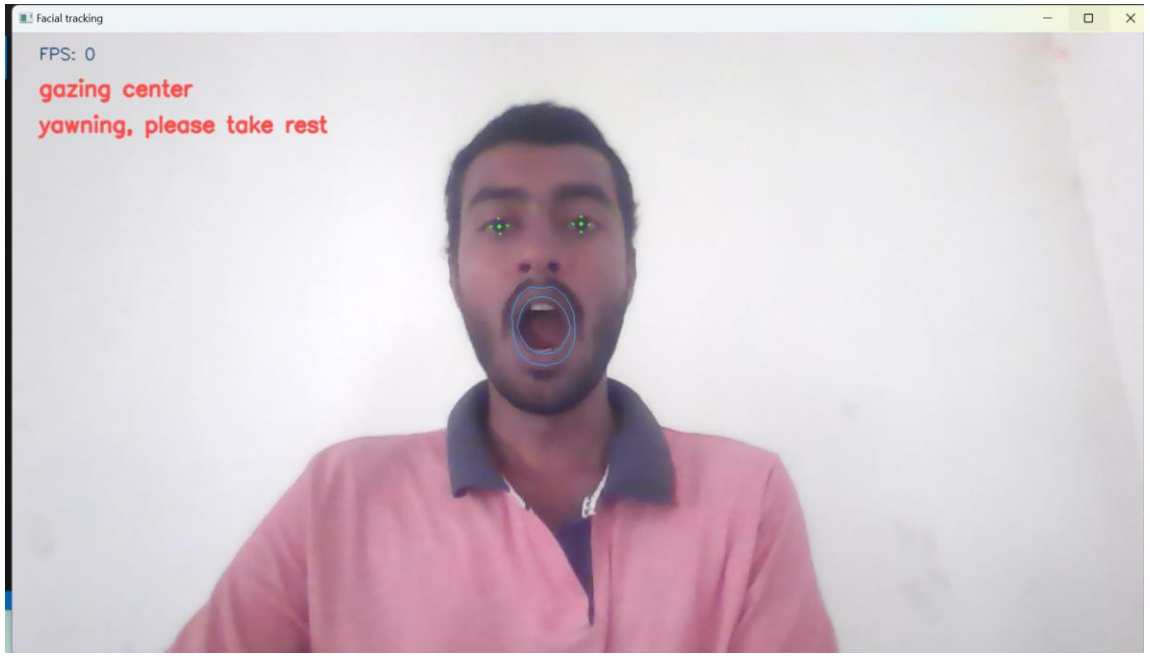


Fig 2: Unsafe Driving action Prediction(Yawning)

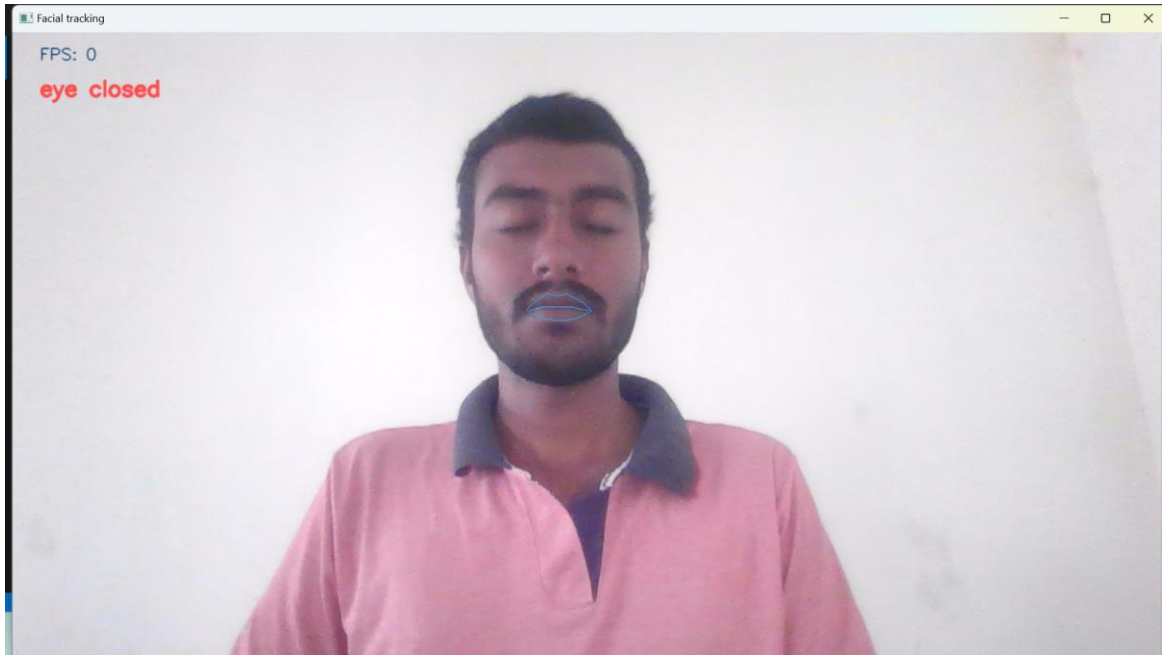


Fig 3: unsafe driver action Prediction (eyes closed , Driver asleep)

CONCLUSION

- Live detection of driver’s behavior is one of the challenging problems faced by automobile companies. In the recent time, as radios, mobile phones and other smart gadgets have become common, accidents occurring due to usage of these devices while driving has increased manifold.
- We investigate the distracted driver’s posture as a part of human action recognition to recognize the driver’s behavior. Our objective is to classify the images as accurately as possible.
- We propose a method where the algorithm classifies different states of the driver and if it senses that the driver is exhibiting unsafe posture, it will ring an alarm and send a message to the driver’s manager.

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Yang Xing, Chen Lv, Huaji Wang, Dongpu Cao, Efstathios Velenis, Fei-Yue Wang,Driver Activity Recognition for Intelligent Vehicles: A Deep Learning Approach. IEEE Transactions on Vehicular Technology, Volume 68, Issue 6, 2019.

3. Distracted driver classification using deep learning Authors: Munif Alotaibi, Bandar Alotaibi. Published Year: Springer Nature on Signal, Image and Video Processing, Vol 14, 617–624, 2020 Methodology: Three advanced deep learning models, namely, the residual network (ResNet), the hierarchical recurrent neural network (HRNN), and the Inception architecture by combining them into one model. This research investigates distracted driver posture recognition as a part of the human action recognition framework. Increasing number of time.

4. Detecting Human Driver Inattentive and Aggressive Driving Behavior using Deep Learning Authors: Monagih. Alkinani, Wazir Zada khan and Quratulain Arshad. Published Year: IEEE Access, vol. 8, pp. 105008-105030, 2020. Methodology: In this system they basically looked upon into two major categories; Distraction and Fatigue/Drowsiness. Deep learning based systems such as CNN, RNN was used. A minor alteration into the input images lead to completely wrong results with a less accuracy rate. Real time processing of heterogeneous data might result in processing delays.

5 .Driver Behavior Recognition via Interwoven Deep Convolutional Neural Nets with Multi-Stream Inputs. Authors: Chaoyun Zhang, Rui Li, Woojin Kim, Daesub Yoon and Paul Patras. Published Year: IEEE Access, vol. 8, pp. 191138-191151, 2020. Methodology: An original driver behavior recognition system that can classify user actions accurately in realtime using input from in-vehicle cameras. The prediction accuracy of InterCNN drops slightly when the inputs are blocked (lossy input). Driver behavior recognition system do not incorporate robust image stabilization software.

THANK YOU